FULL-STACK TECHNICAL SPECIFICATION

DRISHTI-NER Architecture Specification

Frontend, Backend, Database, Hosting & DevOps Architecture Breakdown

Operational Scope: 8 North Eastern States (Assam, Arunachal, Meghalaya, Manipur, Mizoram, Nagaland, Sikkim, Tripura)

1. FRONTEND	2. BACKEND	3. DATABASE	4. HOSTING	5. DEVOPS
React 19 + Vite 8	**Node.js + Express**	**Local & Cloud Store**	**Google Cloud Run**	**esbuild + TS Linter**
Leaflet GIS & Tailwind v4	Gemini 2.5 AI SDK (CJS)	IndexedDB, GCS Ready	Port 3000, 0.0.0.0 Host	PWA Service Worker

1. Frontend (Client-side): What the user sees & touches

The client-side application delivers an accessible, ultra-responsive single-page GIS cockpit and mobile progressive web app (PWA) engineered for low-connectivity Himalayan and mountainous field conditions.

Technology / Component	Role in Client Experience & Interaction
React 19.0 (SPA)	Declarative concurrent rendering, zero-lag state trees, memoized GIS calculations.
TypeScript (Strict)	End-to-end type safety across sensor telemetry, hazard matrices, and GPS coordinates.
Tailwind CSS v4	High-contrast light & dark palettes, WCAG 2.1 AA accessibility, dynamic type scaling.
Leaflet 1.9.4 GIS	Hardware-accelerated map engine with Survey of India boundary lock & blackout mask.
Motion 12	Smooth drawer animations, situational hazard status ripples, and alert sirens.
Lucide React Icons	Clear, unambiguous visual iconography for road blocks, helplines, and sensors.
Web Speech API (TTS)	Multilingual emergency audio broadcasts across 28 regional North Eastern dialects.
Progressive Web App	Vite PWA service worker offline caching; app installs directly to home screen.

2. Backend (Server-side): Server environment, logic & APIs

- **Runtime & Framework:** Node.js LTS with Express 4.21 web framework running inside Cloud Run containers.
- **Single-Bundle CJS:** Compiled with esbuild into dist/server.cjs with externalized packages for cold starts.
- **Google Gemini 2.5 AI:** Server-side @google/genai SDK proxy for damage assessment & plain-language SITREPs.
- **Telemetry Engine:** Ingests Sentinel-1 InSAR, NASA GPM IMERG 24-hr rain, and NASA SMAP soil moisture.
- **Core API Endpoints:** GET /api/health (heartbeat), POST /api/analyze-incident, POST /api/generate-sitrep.
- **API Security:** Strict server-side containment of GEMINI_API_KEY; zero browser-side secret exposure.

3. Database & Storage: Persistent data & uploaded files

- **Client Storage:** localStorage & IndexedDB: Stores citizen profiles, active emergency alerts, and offline queues.
- **PWA Cache Storage:** Workbox Service Worker caches base map tiles, vector styles, and core JS bundles.
- **Cloud Database (Ready):** Google Cloud Firestore / Firebase SDK configured for synced incident reports & multi-role RBAC.
- **File & Media Storage:** Base64 image serialization for immediate preview; Google Cloud Storage (GCS) bucket bucket-ready for drone photos.
- **GIS Spatial Datasets:** High-precision WGS84 GeoJSON polygons covering 8 NER state boundaries and active fault lines.

4. Hosting & Deployment: Where the app lives & deployment model

- **Serverless Platform:** Google Cloud Run container architecture with automated HTTPS termination & auto-scaling.
- **Networking & Port:** Hardcoded container ingress on Port 3000, bound to host 0.0.0.0 via reverse proxy.
- **Dev Environment:** Cloud Run development container (Live HMR-disabled incremental workspace).
- **Production Staging:** Shared preview sandbox for live multi-device validation across web and mobile.
- **Production Start Script:** node dist/server.cjs serves optimized static client files and proxies backend API routes.

5. DevOps & Tools: Version control, pipelines & build tooling

• **Version Control:**	Standard Git repository architecture; 1-click GitHub export and ZIP archive generation.
• **Vite 8 & esbuild:**	Sub-second client bundle compilation and dual-stage backend CommonJS bundling.
• **Type Checking & Lint:**	tsc --noEmit validation ensuring zero undefined references or missing interface props.
• **Package Ecosystem:**	Strict npm dependency management with zero unvetted runtime dependencies.
• **PDF Build Tool:**	Automated headless jsPDF generator creating certified technical documentation on demand.

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